



8086 Designing

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9.1 8086 System Designing

- Q.** Design 8086 based system with following specifications : **(Dec. 15, 10 Marks)**
- (i) 8086 in minimum mode working at 8MHz.
 - (ii) 32KB EPROM using 16KB devices.
 - (iii) 64KB SRAM using 32KB devices.
- Q.** Design 8086 based minimum mode system for following requirements : **(May 16, 12 Marks)**
- (I) 256 KB of RAM using 64 KB x 8-bit device
 - (II) 128 KB of RAM using 64 KB x 8-bit device
 - (III) Three 8-bit parallel ports using 8255
 - (IV) Support for 8 interrupts
- Q.** Design 8086 based system for following requirements : **(Dec. 16, May 17, 10 Marks)**
- (i) Clock frequency 5 MHz
 - (ii) 512 KB RAM using 32 KB x 8
 - (iii) 256 KB ROM using 32 KB x 8
- Q.** Design 8086 based system with following specifications. **(Dec. 17, 10 Marks)**
- (i) 8086 is working in minimum mode at 10 MHz.
 - (ii) 8KB EPROM using 2KB chips.
 - (iii) 16KB SRAM using 8KB chips.
- Discuss system with memory address map.
- Q.** Design 8086 based system for following specifications : **(Q. 2(b), Dec. 18, 10 Marks)**
- (i) 8086 in minimum mode with clock frequency 5MHz.
 - (ii) 64KB EPROM using 16KB X 8 chips
 - (iii) 16KB RAM using 8KB X 8 chips
- Q.** Design 8086 based system for following specifications : **(May 19, 10 Marks)**
- (i) 8086 in minimum mode with clock frequency 5MHz.
 - (ii) 128 KB EPROM using 32KB*8 chips
 - (iii) 32 KB RAM using 16KB*8 chips



9.1.1 Designing Problem 1

Q. Design an 8086 based Maximum Mode system working at 6 MHz having the following :

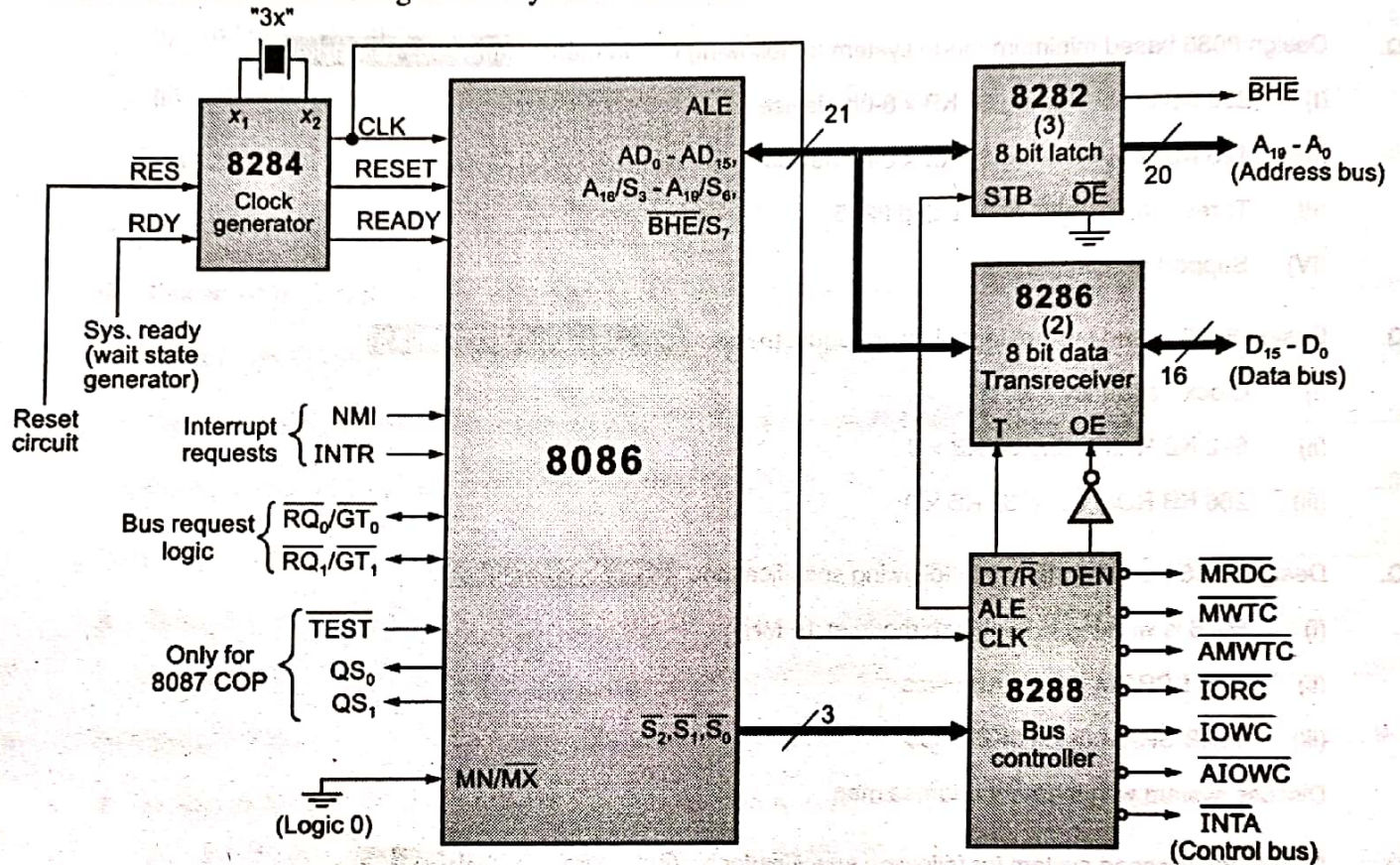
32KB EPROM using 16KB chips, 128KB RAM using 32KB chips,

Two 16-bit input and two 16-bit output ports all interrupt driven

(20 Marks)

Soln. :

Show 8086 max mode config with a crystal of 18 MHz.



(1E1) Fig. 9.1.1 : Maximum mode

Memory Calculations

EPROM

Required = 32 KB, Available = 16 KB

No. of chips = 2 chips.

Starting address of EPROM is calculated as:

FFFFFH – (Space required by total EPROM of 32 KB)

FFFFFH

– 7FFFH

F8000H



Size of a single EPROM chip = 16 KB

$$= 16 \times 1\text{KB} = 2^4 \times 2^{10} = 2^{14}$$

= 14 address lines

= (A14... A1)

RAM

Required = 128 KB, Available = 32 KB

No. of chips = 4 chips.

Starting address of RAM is: 00000H

Size of a single RAM chip = 32 KB

$$= 32 \times 1\text{KB} = 2^5 \times 2^{10} = 2^{15}$$

= 15 address lines

= (A15 ... A1)

Memory Map

Memory Chip	Address Bus																			Memory Address	
	A19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1		A0
RAM 1 (LB)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	00000H
	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	OFFFEH
RAM 2 (HB)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	00001H
	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	OFFFFH
RAM 3 (HB)	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10000H
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1FFFEH
RAM 4 (HB)	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10001H
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1FFFFH
EPORM 1 (LB)	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	F8000H
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	FFFFEH
EPORM 2 (HB)	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	F8001H
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFFFH

(1E2)

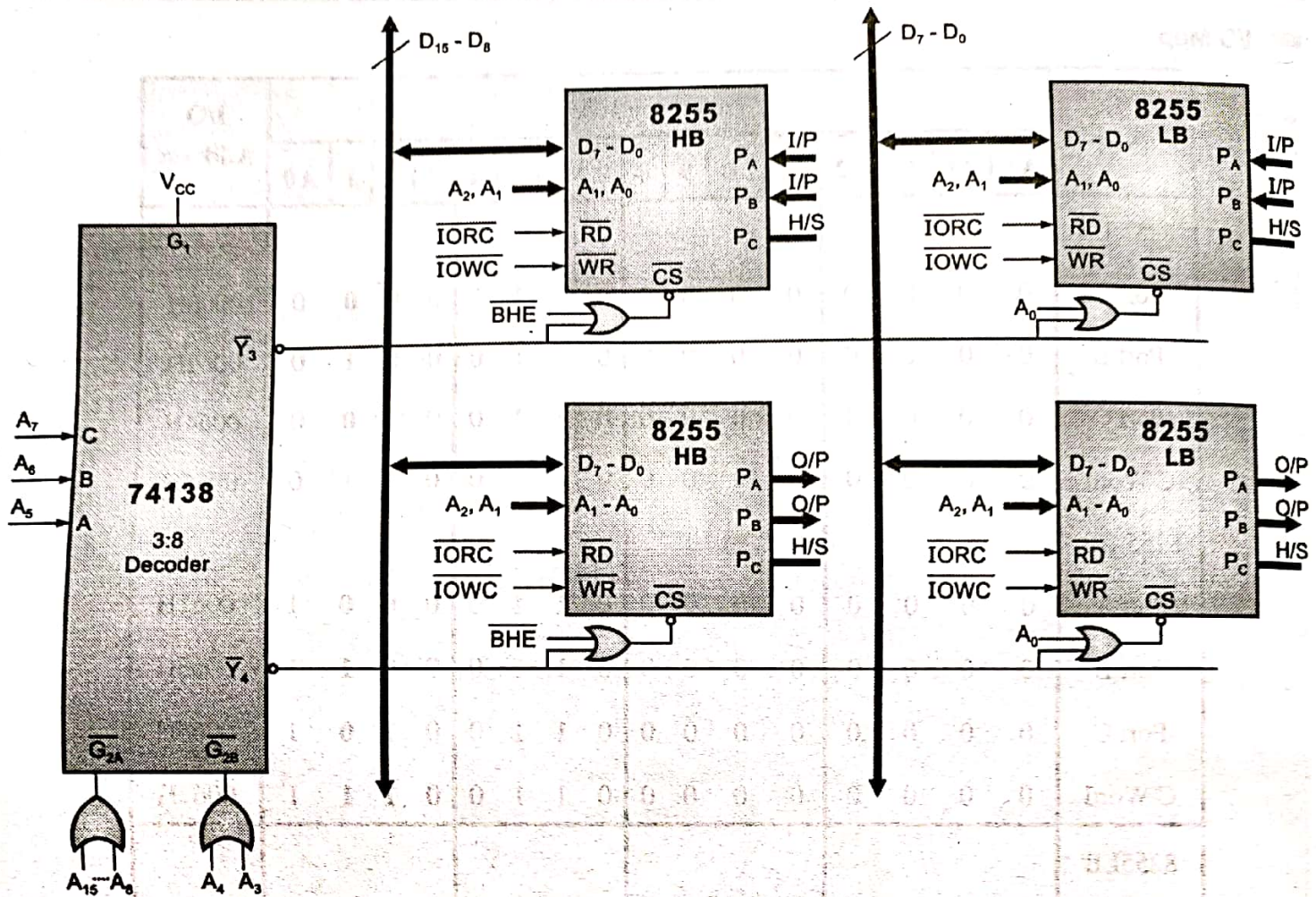




I/O Map

I/O Prot	Address Bus																I/O Address
	A1	14	13	12	11	10	9	8	7	6	5	4	3	2	1	A0	
8255 LB																	
Port A	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0060H
Port B	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0062H
Port C	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0064H
C Word	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	0	0066H
8255 HB																	
Port A	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1	0061H
Port B	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	0063H
Port C	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0065H
C Word	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	0067H
8255LB																	
Port A	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0080H
Port B	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0082H
Port C	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0084H
C Word	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	0086H
8255 HB																	
Port A	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0081H
Port B	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0083H
Port C	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1	0085H
C Word	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	1	0087H

(1E4)



(1E5) Fig. 9.1.3 : I/O diagram

Important points to remember for I/O Designing

- Normally I/O devices are mapped using I/O mapped I/O which means I/O devices are given I/O addresses
- Here I/O addresses can be either 8-bit or 16 bit.
- If the question says direct addressing mode or fixed port addressing, Then use an 8-bit address like 80H (A7-A0).
- If the question says indirect addressing or variable port addressing, Then use 16-bit address like 8000H (A15-A0).
- If nothing is mentioned, use any of the above techniques.
- If memory mapped I/O is asked (Very rare), then remember the following changes

Give the I/O device a 20-bit unused memory address like 80000H (A19-A0)

Connect MEMR# and MEMW# signals to the I/O device instead of the usual IOR# and IOW# signals



9.2 Compare I/O MAPPED I/O and MEMORY MAPPED I/O

Sr. No.	I/O MAPPED I/O	MEMORY MAPPED I/O
1	I/O device is treated as an I/O device and hence given an I/O address.	I/O device is treated like a memory device and hence given a memory address.
2	I/O device has an 8 or 16 bit I/O address.	I/O device has a 20 bit Memory address.
3	I/O device is given IOR# and IOW# control signals	I/O device is given MEMR# and MEMW# control signals
4	Decoding is easier due to lesser address lines	Decoding is more complex due to more address lines
5	Decoding is cheaper	Decoding is more expensive
6	Works faster due to less delays	More gates add more delays hence slower
7	Allows max $2^{16} = 65536$ I/O devices	Allows many more I/O devices as I/O addresses are now 20 bits.
8	I/O devices can only be accessed by IN and OUT instructions.	I/O devices can now be accessed using any memory instruction.
9	ONLY AL/ AH/ AX registers can be used to transfer data with the I/O device.	Any register can be used to transfer data with the I/O device.
10	Popular technique in Microprocessors.	Popular technique in Microcontrollers.

9.3 Designing Problem 2

Q. Design an 8086 based Minimum Mode system working at 6 MHz having the following:
128KB EPROM using 32KB chips, 128KB RAM using 64KB chips

Soln. :

EPROM

Required = _____ KB

Available = _____ KB

No. of chips = _____ chips.

Starting address of EPROM is calculated as:

FFFFFH – (Space required by total EPROM of _____ KB)

F F F F F H

- _____ H

_____ H

Size of a single EPROM chip = _____ KB; = _____ × 1KB;

= $2^{\text{---}} \times 2^{10}$; = $2^{\text{---}}$; = _____ address lines = (A _____ ... A1)

RAM

Required = _____ KB

Available = _____ KB

No. of chips = _____ chips.

Size of a single RAM chip = _____ KB; = _____ × 1KB;

= $2^{\text{---}} \times 2^{10}$; = $2^{\text{---}}$; = _____ address lines = (A _____ ... A1)



Memory Map

Memory	Address Bus																				Memory Address
CHIP	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	A0	
RAM 1 Begin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	00000H
RAM 1 End																					
RAM 2 Begin																					
RAM 2 End																					
EPROM 1 Begin																					
EPROM 1 End																					
EPROM 2 Begin																					
EPROM 2 End																					
EPROM 3 Begin																					
EPROM 3 End																					
EPROM 4 Begin																					
EPROM 4 End	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFFFFH



9.4 Designing Problem 3

- Q. Design an 8086 – 8087 based Maximum Mode system working at 6 MHz having the following:
64 KB EPROM using 16 KB chips, 256 KB RAM using 64 KB chips,

Soln. :

EPROM

Required = _____ KB

Available = _____ KB

No. of chips = _____ chips.

Starting address of EPROM is calculated as:

FFFFFH – (Space required by total EPROM of _____ KB)

F FFFF H

- _____ H

_____ H

Size of a single EPROM chip = _____ KB; = _____ x 1KB;

= $2^{\text{---}} \times 2^{10}$; = $2^{\text{---}}$; = _____ address lines = (A _____ ... A1)

RAM

Required = _____ KB

Available = _____ KB

No. of chips = _____ chips.

Size of a single RAM chip = _____ KB; = _____ x 1KB;

= $2^{\text{---}} \times 2^{10}$; = $2^{\text{---}}$; = _____ address lines = (A _____ ... A1)



Memory Map

Memory	Address Bus																			Memory Address	
CHIP	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	A0	
RAM 1 Begin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
RAM 1 End																					
RAM 2 Begin																					
RAM 2 End																					
RAM 3 Begin																					
RAM 3 End																					
RAM 4 Begin																					
RAM 4 End																					
EPROM 1 Begin																					
EPROM 1 End																					
EPROM 2 Begin																					
EPROM 2 End																					
EPROM 2 Begin																					
EPROM 3 End																					
EPROM 4 Begin																					
EPROM 4 End	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	



9.5 Designing Problem 4

Q. Design an 8086 based Minimum Mode system working at 10 MHz having the following:
16 KB EPROM using 4 KB chips, 64 KB RAM using 8 KB chips,

Soln. :

EPROM

Required = 16 KB

Available = 4 KB

No. of chips = 4 chips.

Starting address of EPROM is calculated as:

FFFFFH – (Space required by total EPROM of 16 KB)

F	F	F	F	F	H
-	3	F	F	F	H
F	C	0	0	0	H

Size of a single EPROM chip = 4KB; = 4 × 1KB;

= $2^2 \times 2^{10}$; = 2^{12} ; = 12 address lines = (A12... A1)

RAM

Required = 64 KB

Available = 8 KB

No. of chips = 8 chips.

Size of a single RAM chip = 8KB; = 8 × 1KB;

= $2^3 \times 2^{10}$; = 2^{13} ; = 13 address lines = (A13... A1)



Memory Map

Memory CHIP	Address Bus																				Memory Address
	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	A0	
RAM 1 Begin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	00000H
RAM 1 End	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	03FFEH
RAM 2 Begin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	00001H
RAM 2 End	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	03FFFH
RAM 3 Begin	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	04000H
RAM 3 End	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	07FFEH
RAM 4 Begin	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	04001H
RAM 4 End	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	07FFFH
RAM 5 Begin	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	08000H
RAM 5 End	0	0	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0BFFEH
RAM 6 Begin	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	08001H
RAM 6 End	0	0	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0BFFFH



Memory CHIP	Address Bus																				Memory Address
	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	A0	
RAM 7 Begin	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0C000H
RAM 7 End	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0FFFEH
RAM 8 Begin	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0C001H
RAM 8 End	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0FFFFH
EPROM 1 Begin	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	FC000H
EPROM 1 End	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	0	FDFFEH
EPROM 2 Begin	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	FC001H
EPROM 2 End	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	FDFFFH
EPROM 3 Begin	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	FE000H
EPROM 3 End	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	FFFFEH
EPROM 4 Begin	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	FE001H
EPROM 4 End	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFFFFH

(1E8)